

切削参数参考表

加工材料 Work Material		高速钢 High Speed Tool Steels SKH (~65HRC)				淬火钢 Hardened Steels SKD11 (~62HRC)				淬火钢 Hardened Steels SKD61 · STAVAX · HPM-38 (~52HRC)			
球头半径(R) Radius	颈长(ℓ1) Under Neck Length	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
0.4	2	30,000	1,200	0.05	0.1	40,000	1,600	0.07	0.1	40,000	2,000	0.1	0.15
	3	30,000	1,200	0.05	0.05	40,000	1,600	0.07	0.1	40,000	2,000	0.1	0.15
	4	30,000	860	0.03	0.05	40,000	1,200	0.05	0.05	40,000	1,600	0.05	0.1
	6	25,000	560	0.01	0.02	30,000	760	0.02	0.03	30,000	1,200	0.03	0.05
	8	25,000	480	0.005	0.01	30,000	600	0.007	0.01	30,000	820	0.01	0.02
0.45	10	20,000	320	0.002	0.003	25,000	380	0.003	0.003	25,000	450	0.005	0.005
	2	30,000	1,300	0.06	0.1	30,000	1,800	0.08	0.15	40,000	2,200	0.1	0.2
	4	25,000	900	0.03	0.05	30,000	1,400	0.04	0.08	40,000	1,800	0.05	0.12
	6	20,000	600	0.015	0.025	25,000	800	0.025	0.035	30,000	1,200	0.035	0.05
	8	20,000	500	0.008	0.015	23,000	700	0.015	0.025	30,000	1,000	0.025	0.04
0.5	2	25,000	1,400	0.08	0.1	30,000	2,000	0.1	0.2	40,000	2,500	0.1	0.3
	3	25,000	1,400	0.08	0.1	30,000	2,000	0.1	0.2	40,000	2,500	0.1	0.3
	4	25,000	1,200	0.05	0.1	30,000	1,800	0.05	0.15	40,000	2,500	0.1	0.2
	5	20,000	920	0.03	0.05	25,000	1,600	0.04	0.1	30,000	2,000	0.05	0.15
	6	20,000	740	0.02	0.05	25,000	1,200	0.04	0.05	30,000	1,800	0.05	0.1
0.6	8	20,000	560	0.02	0.03	25,000	860	0.03	0.04	30,000	1,000	0.04	0.06
	10	18,000	450	0.01	0.02	20,000	620	0.02	0.03	25,000	750	0.03	0.05
	12	16,000	400	0.005	0.01	18,000	520	0.007	0.02	20,000	600	0.01	0.03
	16	14,000	200	0.002	0.003	16,000	250	0.003	0.005	18,000	300	0.005	0.005
	4	25,000	1,600	0.05	0.1	30,000	2,000	0.07	0.2	30,000	2,500	0.1	0.2
0.75	6	20,000	1,200	0.03	0.07	25,000	1,600	0.05	0.1	30,000	2,000	0.07	0.1
	8	20,000	920	0.02	0.05	25,000	1,200	0.03	0.07	30,000	1,600	0.05	0.1
	10	18,000	680	0.01	0.03	20,000	860	0.02	0.05	20,000	1,200	0.03	0.07
	12	18,000	480	0.007	0.02	20,000	620	0.01	0.03	20,000	860	0.02	0.05
	16	14,000	130	0.003	0.007	16,000	250	0.005	0.01	16,000	350	0.01	0.02
0.8	4	25,000	2,000	0.1	0.2	30,000	2,500	0.1	0.3	30,000	3,000	0.15	0.3
	6	25,000	1,600	0.1	0.1	30,000	2,000	0.1	0.2	30,000	3,000	0.15	0.2
	8	20,000	1,200	0.05	0.1	25,000	1,600	0.05	0.2	25,000	2,500	0.1	0.2
	10	20,000	860	0.05	0.05	25,000	1,200	0.05	0.1	25,000	2,500	0.1	0.1
	12	18,000	780	0.02	0.05	20,000	920	0.03	0.1	20,000	1,800	0.05	0.1
0.9	16	16,000	580	0.01	0.02	18,000	650	0.02	0.03	18,000	720	0.03	0.05
	4	18,000	1800	0.08	0.15	20,000	2500	0.1	0.2	25,000	2800	0.15	0.3
	8	18,000	1,600	0.05	0.1	20,000	2,000	0.07	0.15	25,000	2,500	0.1	0.2
	12	14,000	1,200	0.03	0.05	16,000	1,500	0.05	0.07	20,000	1,800	0.07	0.1
	16	12,000	480	0.015	0.02	14,000	600	0.02	0.03	16,000	720	0.03	0.05
1	20	10,000	300	0.01	0.01	12,000	380	0.01	0.02	14,000	500	0.01	0.03
	4	18,000	1800	0.08	0.15	20,000	2500	0.1	0.2	25,000	3000	0.15	0.3
	8	18,000	1600	0.05	0.1	20,000	2000	0.07	0.15	25,000	2500	0.1	0.2
	12	14,000	1200	0.03	0.05	16,000	1000	0.05	0.07	20,000	1800	0.07	0.1
	16	12,000	500	0.015	0.02	14,000	800	0.02	0.03	16,000	720	0.03	0.05
1.25	20	10,000	300	0.01	0.01	12,000	400	0.01	0.02	14,000	500	0.01	0.03
	4	20,000	2,000	0.15	0.3	25,000	2,500	0.2	0.5	25,000	3,000	0.2	0.5
	6	20,000	1,600	0.15	0.3	25,000	2,000	0.2	0.3	25,000	2,500	0.2	0.5
	8	16,000	1,200	0.1	0.2	18,000	1,600	0.1	0.2	20,000	2,000	0.2	0.3
	10	14,000	1,200	0.1	0.1	16,000	1,600	0.1	0.2	18,000	2,000	0.1	0.3
1.5	12	12,000	940	0.05	0.1	14,000	1,200	0.1	0.1	16,000	1,600	0.1	0.2
	16	12,000	940	0.03	0.07	14,000	1,200	0.05	0.08	16,000	1,600	0.07	0.15
	20	10,000	720	0.02	0.03	12,000	820	0.03	0.05	14,000	1,000	0.05	0.1
	25	8,500	420	0.01	0.02	10,000	560	0.02	0.03	12,000	680	0.03	0.05
	30	8,500	240	0.008	0.01	10,000	300	0.01	0.02	12,000	360	0.02	0.03
2.0	6	18,000	2,000	0.15	0.4	20,000	2,300	0.2	0.5	20,000	2,800	0.3	0.4
	8	18,000	1,800	0.12	0.25	20,000	2,100	0.15	0.3	20,000	2,600	0.25	0.3
	12	16,000	1,400	0.08	0.13	18,000	1,800	0.13	0.18	19,000	2,300	0.15	0.25
	16	12,000	1,100	0.04	0.08	15,000	14,000	0.06	0.13	1,700	1,750	0.1	0.2
	20	10,000	1,000	0.03	0.05	14,000	1,200	0.05	0.1	16,000	1,500	0.07	0.15

切削参数参考表

加工材料 Work Material		高速钢 High Speed Tool Steels SKH (~5HRC)				淬火钢 Hardened Steels SKD11 (~62HRC)				淬火钢 Hardened Steels SKD61 · STAVAX · HPM-38 (~52HRC)			
球头半径(R) Radius	颈长(l1) Under Neck Length	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
1.5	6	14,000	2,000	0.2	0.5	18,000	2,500	0.2	0.6	20,000	3,000	0.2	0.8
	8	14,000	2,000	0.2	0.5	18,000	2,500	0.2	0.6	20,000	3,000	0.2	0.8
	10	14,000	1,500	0.1	0.3	18,000	2,000	0.2	0.4	20,000	2,500	0.2	0.6
	12	14,000	1,500	0.1	0.3	18,000	2,000	0.2	0.4	20,000	2,500	0.2	0.6
	16	12,000	1,200	0.1	0.2	16,000	1,600	0.1	0.3	18,000	2,000	0.1	0.4
	20	12,000	960	0.1	0.1	16,000	1,200	0.1	0.2	18,000	1,600	0.1	0.3
	25	10,000	800	0.05	0.07	14,000	920	0.07	0.15	16,000	1,200	0.1	0.2
1.75	30	8,600	600	0.03	0.05	12,000	640	0.05	0.07	14,000	750	0.07	0.1
	10	14,000	1500	0.15	0.5	16,000	2300	0.2	0.6	20,000	3000	0.3	1.2
	15	14,000	1,500	0.13	0.4	16,000	2,300	0.15	0.5	20,000	3,000	0.25	1
	20	12,000	1,200	0.1	0.2	15,000	1,800	0.1	0.3	18,000	2,500	0.18	0.6
	25	10,000	1,000	0.06	0.12	14,000	1,600	0.1	0.2	16,000	1,800	0.12	0.35
2	30	9,000	800	0.05	0.08	11,000	950	0.07	0.15	14,000	1,500	0.1	0.25
	12	12,000	2,000	0.2	0.6	16,000	2,500	0.2	0.8	20,000	3,000	0.3	1.5
	16	11,000	1,500	0.15	0.5	15,000	19,000	0.15	0.7	1,800	2,600	0.25	1.2
	20	10,000	1,400	0.1	0.4	14,000	1,800	0.1	0.6	16,000	2,400	0.2	1
	25	10,000	1,000	0.1	0.2	14,000	1,200	0.1	0.4	16,000	1,600	0.2	0.8
30	8,200	1,000	0.05	0.15	10,000	1,200	0.07	0.2	14,000	1,600	0.1	0.3	
	备注 Notes												
※ 切深量的ap表示轴向切入量，ae表示步距量。 ※ 建议使用油雾冷却方式。 ※ 请以相同的比率调整主轴转速和进给速度。 ※ 加工参数会因切深量和机床刚性的状况而有所不同。请每次调整后再使用。 ※ 请根据需要控制刀具的伸出量。 ※ Depth of Cut: ap=Axial Depth of Cut / ae=Radial Depth of Cut. ※ We recommend using oil mist coolant. ※ Adjust both spindle speed and feed at the same rate. ※ Adjust milling conditions according to the volume of depth of cut and rigidity of machine. ※ Length of tool overhang must be as short as possible.													